TITANIC SURVIVAL ANALYSIS

FINAL PROJECT REPORT | Data Analytics & Business Intelligence

A source-grounded study of passenger survival, demographic disparities, travel characteristics, and interactive dashboard design.

Project component	Details
Dataset	Titanic passenger dataset, 891 records
Analysis environment	Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy
Dashboard environment	Microsoft Power BI
Source artifacts	titanic_analysis.ipynb and 1.TSAD.pbix
Report scope	Descriptive analysis, statistical testing, dashboard review, findings, limitations

Prepared from the uploaded project files. All numerical findings cited as results originate in the notebook outputs; dashboard elements are described from the PBIX report layout. No additional raw passenger data or Power BI rendered screenshots were supplied.

Executive snapshot

The notebook records 342 survivors and 549 non-survivors among 891 passengers. The survival rate is 38.38%. Survival varied markedly by sex and passenger class, with female passengers and first-class passengers showing higher observed rates. These patterns represent associations in this dataset, not proof of causation.

1. Executive Summary and Project Objectives

The Titanic Survival Analysis project examines historical passenger-level records to understand which observed passenger characteristics were associated with survival. The work follows a practical analytics workflow: inspect raw data, identify and address missing values, construct explanatory features, perform exploratory and statistical analyses, and communicate findings through a Power BI dashboard.

Primary objectives

1. Quantify overall survival and non-survival. 2. Compare survival across sex and passenger class. 3. Explore age, embarkation port, ticket fare, and family size. 4. Apply basic statistical testing to a prominent association. 5. Present a usable summary of results through interactive business-intelligence visuals.

Key findings

Indicator	Notebook result
Passengers	891
Survivors	342
Non-survivors	549
Survival rate	38.38%
Female survival rate	74.20%
Male survival rate	18.89%
First-class survival rate	62.96%
Third-class survival rate	24.24%

The difference between female and male survival rates is 55.31 percentage points. The difference between first-class and third-class survival rates is 38.72 percentage points. Both gaps are substantial in descriptive terms; interpretation should account for overlap between demographic and socioeconomic factors.

2. Dataset Description and Initial Audit

The notebook loads a CSV file named titanic.csv and reports a dataset shape of 891 rows and 12 original columns. The outcome field, Survived, is binary: 1 indicates survival and 0 indicates non-survival. Each row corresponds to one passenger record.

Field(s)	Analytical role
PassengerId, Name, Ticket	Identifiers and passenger descriptors
Survived	Binary outcome variable
Pclass, Fare	Travel class and ticket fare
Sex, Age	Passenger demographics
SibSp, Parch	Family relationships aboard
Cabin, Embarked	Cabin reference and port of embarkation

Missing-value audit

Column	Missing records	Share of 891
Age	177	19.87%
Cabin	687	77.10%
Embarked	2	0.22%
Other original columns	0	0%

The descriptive output gives a mean observed age of 29.70 years before imputation, a median observed age of 28 years, a minimum age of 0.42 years, and a maximum of 80 years. Fare has a mean of 32.20 and median of 14.45, indicating a distribution influenced by comparatively expensive tickets. The maximum recorded fare is 512.33.

The notebook calls df.duplicated() and displays a boolean series with sampled rows marked False. It does not show an aggregate duplicate count, so a claim that all duplicate records were eliminated would not be supported.

3. Data Cleaning and Feature Engineering

Handling incomplete values

Age missing values are replaced with the column mean; Embarked missing values are filled using the most frequent category; Cabin missing values are replaced with the string Unknown. The notebook then confirms zero missing values across the resulting columns. This retains all 891 passenger records rather than dropping passengers with incomplete cabins.

Engineered variables

Feature	Definition and intended use
FamilySize	SibSp + Parch + 1; total recorded traveling family group size including passenger
IsAlone	1 when FamilySize equals 1, otherwise 0
Title	Honorific extracted from the passenger Name field
CabinKnown	Intended cabin-availability flag; implementation issue described below

Important quality-control issue

The CabinKnown flag is calculated after missing Cabin values have already been filled with Unknown. Because Unknown is not a null value, the expression df["Cabin"].notna().astype(int) sets CabinKnown to 1 for every passenger. It therefore does not represent whether an original cabin was known. A corrected approach is to create the indicator before filling nulls, or define it as (Cabin != "Unknown").astype(int). Any dashboard or model using the existing flag should be reviewed.

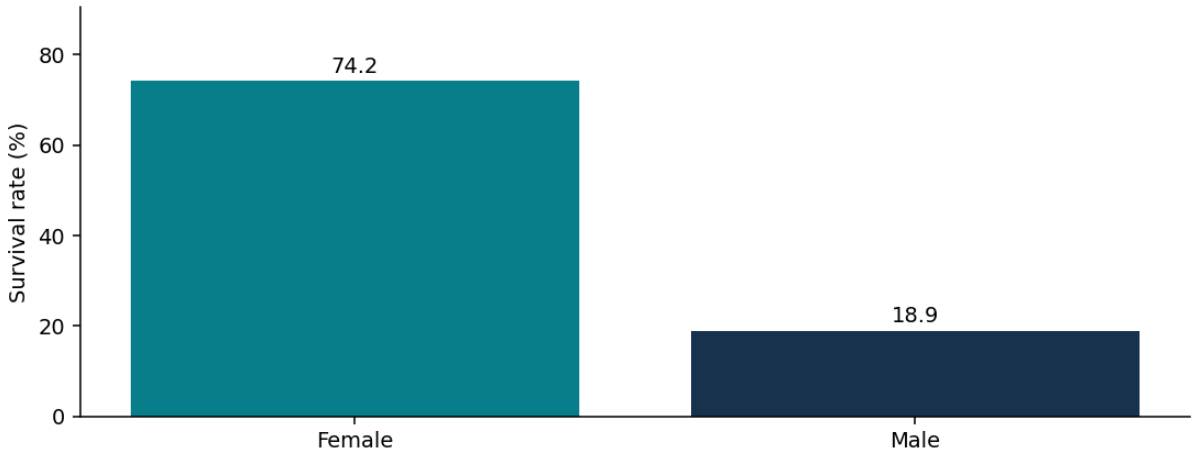
Export and reproducibility

The notebook reports an updated dataset of 891 rows and 16 columns. It saves the CSV to data for analysis/titanic_updated.csv but prints a location under csv/titanic_updated.csv; the two paths do not match. The analysis code uses a Windows-specific absolute input path, which should be replaced by a project-relative path for easier reproduction.

Mean imputation preserves dataset size but compresses age variation, influences age-based groups, and may bias comparisons involving age. These results should therefore be described as based on imputed age values where applicable.

4. Overall Survival and Gender-Based Findings

The notebook reports 342 survivors (38.38%) and 549 non-survivors (61.62%). Non-survivors outnumber survivors by 207 records. This baseline is important for interpreting subgroup results and any future classification model.



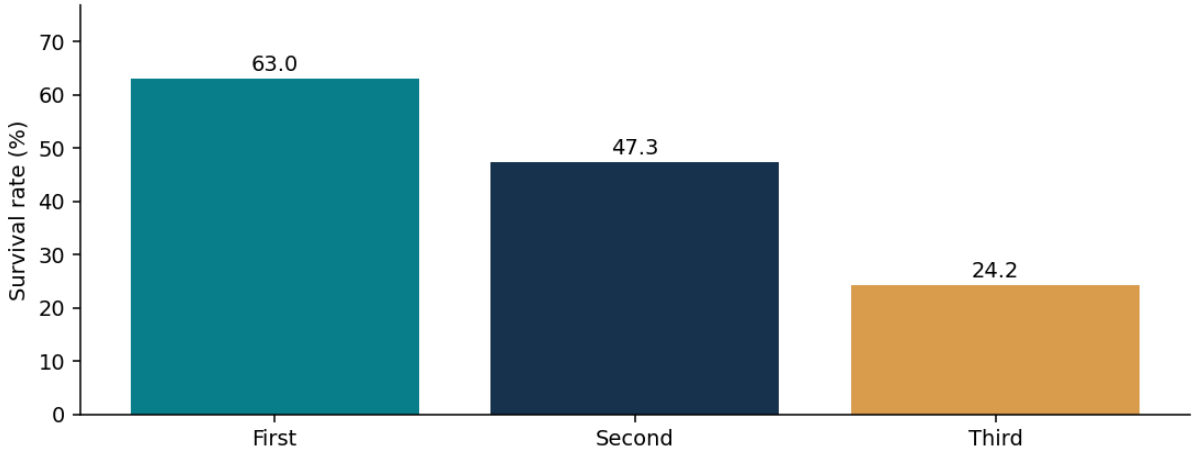
Sex	Passengers	Survivors	Survival rate
Female	314	233	74.20%
Male	577	109	18.89%

Female passengers were almost four times as likely to survive in the observed dataset when comparing subgroup survival proportions (74.20% versus 18.89%). Of the 342 survivors, 233 were women and 109 were men. Conversely, the dataset contains 81 female non-survivors and 468 male non-survivors.

Statistical test

A chi-square test of independence was performed on Sex and Survived. The notebook outputs chi-square = 260.717, 1 degree of freedom, and p = 1.1973570627755645 × 10^-58. Under the usual assumptions of the chi-square test, this is strong evidence that sex and survival are not independent in the observed records. It does not establish a causal mechanism or control for class, age, or other factors.

5. Passenger Class, Age and Survival



Class	Passengers	Survivors	Survival rate
First	216	136	62.96%
Second	184	87	47.28%
Third	491	119	24.24%

Survival rates declined steadily from first to third class. First-class passengers had an observed survival rate about 2.6 times that of third-class passengers. Third class contained 491 passengers, the largest group, and 372 non-survivors. These descriptive differences are consistent with unequal outcomes across travel classes, but the notebook does not isolate class from demographic composition or evacuation access.

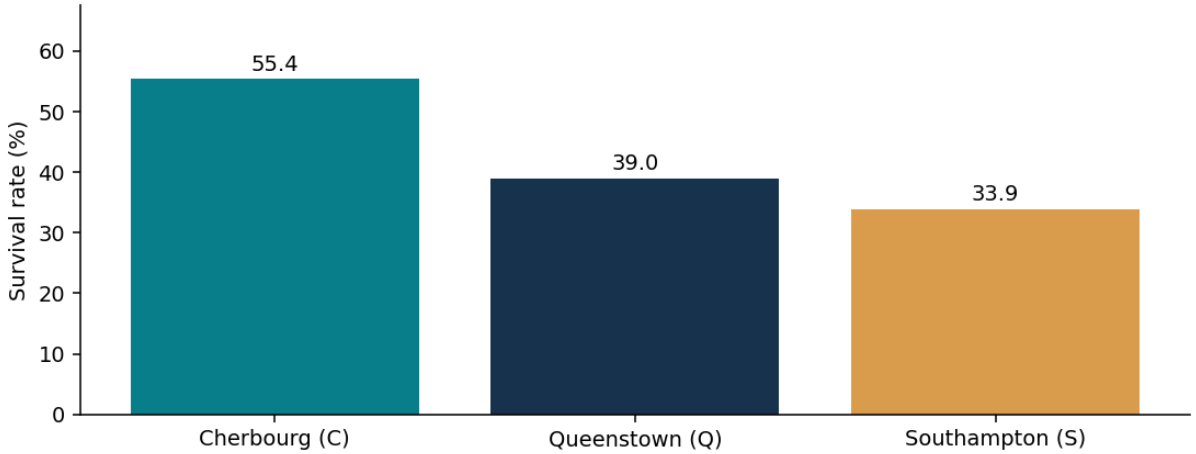
Age comparison

Outcome	Mean age	Median age	Minimum	Maximum
Non-survivors	30.42	29.70	1.00	74.00
Survivors	28.55	29.70	0.42	80.00

Survivors had a mean age about 1.87 years lower than non-survivors after missing-age imputation. The median age is identical across the two groups because missing ages were replaced by the overall mean of approximately 29.70. These summary statistics do not establish that children consistently survived more often: the notebook does not output a grouped age-band survival table.

The Power BI layout includes an Age Group slicer, indicating that age categories are used in dashboard interaction. The PBIX report layout does not itself provide category boundaries or aggregated values, so precise age-group findings cannot be verified from the supplied files.

6. Embarkation Port and Ticket Fare



Port code	Passengers	Survival rate
C	168	55.4%
Q	77	39.0%
S	646	33.9%

Passengers embarking at Cherbourg had the highest observed survival proportion, followed by Queenstown and Southampton. Southampton accounted for the largest number of passengers. Two missing embarkation entries were imputed with the mode before analysis. The observed differences may reflect class and demographic composition rather than an independent effect of embarkation location.

Ticket fare findings

Outcome	Mean fare	Median fare
Survivors	48.40	26.00
Non-survivors	22.12	10.50

Survivors paid higher fares on average, with a mean difference of 26.28 fare units and a median difference of 15.50. The notebook uses a box plot to compare fare distributions. Given the large spread in fare values, median comparisons are especially informative. Fare and class are likely related, so the observed fare association should not be interpreted as evidence that fare independently determined survival.

7. Family Size, Correlation and Statistical Interpretation

Family size	Passengers	Survival rate
1	537	30.4%
2	161	55.3%
3	102	57.8%
4	29	72.4%
5	15	20.0%
6	22	13.6%
7	12	33.3%
8	6	0.0%
11	7	0.0%

Passengers traveling alone, defined by FamilySize = 1, accounted for 537 of 891 records and had a 30.4% survival rate. Groups of two, three, and four recorded higher rates, peaking at 72.4% for family size four. Larger family groups often had lower observed survival, including zero survivors in the small family-size-eight and family-size-eleven categories. These extreme categories have very small samples and should not be generalized without caution.

Correlation heatmap

The notebook constructs a correlation matrix of numeric fields and visualizes it with a Seaborn heatmap. The notebook output stores the figure but does not provide a printed matrix of coefficients in the extracted results. Consequently, this report does not invent specific correlation values. In a revised project, selected correlations should be printed and interpreted alongside the visual.

What the statistical evidence establishes

The chi-square result supports a strong association between sex and survival. The class, fare, port, age, and family-size results are descriptive subgroup comparisons. No multivariable regression, confidence intervals, predictive accuracy, or causal analysis are shown in the uploaded notebook. Claims about independent predictors or model performance would therefore go beyond the supplied evidence.

8. Power BI Dashboard Review and Recommendations

The uploaded PBIX report layout contains one report page labeled Page 1 with 14 visual containers. The underlying dataset is referenced as titanic_updated. The page contains a text box, five card visuals, a donut chart, multiple column/bar charts, and two slicers. The layout identifies measures including Total Passengers, Total Survivors, Total Deaths, and Survival Rate, and uses fields including Sex, Pclass, and Age Group.

Dashboard communication

The card visuals are suitable for presenting high-level counts and rates, while the donut chart and clustered charts provide breakdowns across passenger groups. A Sex slicer and an Age Group slicer allow viewers to explore filtered subsets. These features complement the notebook by moving from static plots to interactive exploration.

Recommended narrative for the final presentation

Start with the total passenger count and overall survival rate. Next, show the pronounced sex difference and the progressive decline in survival from first to third class. Use age filtering to explore demographic patterns, and discuss fare, embarkation, and family size as supporting context. Close by explaining why descriptive associations should not be interpreted as causal conclusions.

Dashboard limitations

The PBIX file stores the report definition and data model, but this report did not execute Power BI or capture the live rendered visuals. Therefore, the presence and types of visuals can be verified from the layout, while displayed KPI values, visual formatting, exact age-bin rules, and interactive filter behavior were not independently tested. Dashboard values should be checked against notebook totals after refresh.

Suggested improvements

Add explanatory subtitles stating the denominator of each survival percentage. Distinguish survival count from survival rate in titles. Display sample sizes for small family groups, add a data-quality note on imputed ages, verify DAX calculations, and correct the CabinKnown flag before using it in any report or model.

9. Conclusion, Limitations and Future Scope

Conclusion

The Titanic Survival Analysis project demonstrates an end-to-end descriptive analytics workflow on 891 passenger records. Of those passengers, 342 survived, giving an overall survival rate of 38.38%. The strongest visible contrast is by sex: 74.20% of female passengers survived compared with 18.89% of male passengers. Passenger class also shows a large gradient, from 62.96% survival in first class to 24.24% in third class. Fare, embarkation port, and family size show further differences in observed outcomes.

The chi-square test provides formal statistical evidence of association between sex and survival. The Power BI dashboard adds a presentation layer through summary cards, comparative visuals, and demographic slicers. Together, the notebook and dashboard provide a useful portfolio demonstration of cleaning, feature engineering, EDA, statistical reasoning, and visual reporting.

Study limitations

The dataset is a sample of Titanic passenger records, not necessarily a complete historical manifest. Missing Age values were mean-imputed, Cabin was predominantly unknown, and the existing CabinKnown indicator is incorrect. Several subgroup categories contain few passengers. The analysis is observational and mostly unadjusted; it cannot prove causality or quantify independent effects. The dashboard was inspected structurally rather than rendered and interactively validated.

Future scope

Future development could include corrected cabin-availability features, age-group tables with clearly documented boundaries, class-by-sex comparisons, uncertainty intervals, logistic regression for adjusted associations, and predictive models evaluated with stratified train/test splits and appropriate metrics. Any predictive work should prevent data leakage by fitting imputation and transformations on training data only.

Source and verification notes

Primary sources: uploaded titanic_analysis.ipynb (47 cells; outputs for data inspection, EDA and chi-square test) and 1.TSAD.pbix (one report page and 14 visual containers). No external sources were used to generate the findings. Numeric rates in this report reflect the displayed notebook outputs, with a few simple derived differences and rounded percentages. This document should be reviewed against the refreshed dashboard before final submission.